

Monitoring

Monitoring in an azure landing zone is continuous collection analysis and alerting of the whole system to ensure azure resources are available, healthy, secure, performing, cost efficient, and compliant.

So when we monitor end to end system, we will consider **Infrastructure monitoring**(VM, VMSS, AKS, SQL, Storage, app services, functions, logic App, Web App)

Networking monitoring (Vnet/Subnet Flow Logs, Firewall logs, Express Route, VPN, Loadbalancer, application gateway, frontdoor)

Identity Monitoring (Entra ID logs, MFA logs, PIM logs, RBAC logs, conditional RBAC logs)

Security Monitoring (Integration of defender to different services)

Backup and DR logs (Backup Failed or not, Site Recovery Failed or not)

Cost Optimization Tracking, Application Level Logging

For Monitoring we have two solutions, Azure Native Monitoring and Open Source

Azure Native

Azure Monitor - The AIM of Azure Monitor is to create central logging system which collects metrics, logs, events and alerts.

For collection or storing the logs we have log analytics workspace which stores logs in column and table format.

Example: It stores AKS logs, VM logs, VMSS logs, Firewall Logs etc...

once the logs will be collected everything is being queried with KQL ie kusto query language

For Application, We have application insights, which tracks failures, response times, performance, request rate, user sessions of the application

Open Source

Every monitoring tool collects Metrics(CPU, Memory, Disk, Latency, Availability) and Logs(Activity Logs, Subscription logs, Diagnostic Logs).

Every monitoring tool works in similar fashion. monitoring agent will be installed on the resources which is required to monitor. in Azure monitor that agent is called AMA(Azure Monitor Agent) that is responsible for collecting windows events, linux syslogs, performance counters, custom logs using DCR(Data Collection Rules). It defines what to collect. Either to collect SysLog, to event logs



3. Log sent to Log Analytics Workspace



1. Install Azure Monitor Agent

2. DCR Collection Rule

4. Run KQL Query to extract data

5. Set Alert Rules

6. Alert should be sent to specific emails, action group will be created for the same

Infra Monitoring: CPU, Memory, Disk, Availability, Heartbeat, Process, OS Events, Services

Example: CPU > 70% - Alerts Rule

More than 3 failed logins - Alert Rule

With the KQL query reports, we can pin that reports to azure dashboards, Azure Managed grafana, Power BI so Landing Zone Dashboard Include Resource Health

Best practices for Azure monitoring

Always develop centralized logging system in dedicated log analytics workspace.

Always configure DCR collection Rule for pulling most critical logs instead of pushing all logs to workspace.

Enable diagnostic settings for all critical resources.

Always create action group, alerts integrated with teams, slack, Email or any other integration platform.

Create Dashboard for critical resources(Health and Cost).

Set Alerts for Budget and Cost.

Always Monitor Backup, DR, Compliance, Governance, Cost, Resource health, Continuously for all resources.